MIL-P-10455A
18 APRIL 1951
SUPERSEDING
MIL-P-10455(CmlC)
22 August 1950

MILITARY SPECIFICATION

PHOSGENE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers phosgene for use in munitions and chemical manufacture.

1.2 Classification.—Phosgene shall be of the following grades as specified (see 6.1):

Grade A—For use in munitions, maximum free chlorine 1.0 percent.

Grade B—For use in chemical manufacture, maximum free chlorine 0.05 percent.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications.—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

MILITARY SPECIFICATION

MIL-C-3250—Container, One-Ton, Type D.

U. S. ARMY SPECIFICATION

94-40645 —Marking; Exterior, Domestic and Export Shipment by Contractors.

96-31-64 —Container, One-Ton, Type E.

100-2 —Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.

(*Army.*—Copies of specifications may be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Military and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Other publications. — The following publications, of the issue in effect on date of invitation for bids, form a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y.)

INTERSTATE COMMERCE COMMISSION REGULATIONS

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

FSC 1365

905080—51—(671)

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(Information as to the availability of Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., may be obtained from the Interstate Commerce Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 The material shall conform to the requirements shown in table I.

TABLE I.—*Chemical and physical requirements*

Requirements	*Maximum*	*Minimum*
	Percent	*Percent*
Purity (percent phosgene)	---	98.0
Acidity (calculated as HCl)	0.50	---
Residue on evaporation	0.50	---
Free chlorine:		
Grade A	1.00	---
Grade B	0.05	---

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Lot.**—The contents of each storage tank of finished phosgene in the manufacturing unit shall constitute a lot.

4.2 **Sampling.** — Pass sufficient phosgene from the storage tank to purge the line from which sample is taken. Then pass approximately one liter of phosgene from the tank into a steel sampling cylinder of suitable size. Label the cylinder so as to insure the identity of the sample with the lot represented (see 6.2 and 6.4).

4.3 **Inspection.**

4.3.1 *Procedures.*—For Naval purchases, general inspection shall be in accordance with General Specifications for Inspection of Material.

4.3.2 *Packing and marking.*—The inspector shall inspect the packing and marking of the material for conformance with section 5 (see 6.3).

4.4 **Tests.**

4.4.1 *Reagents.*

4.4.1.1 *General.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and corrections shall be applied when significant.

4.4.1.2 *Acetone, dry.* — Place 1700 ml. of acetone conforming to American Chemical Society Specification (see 6.5) in a 2-liter flask. Add 100 gm. of anhydrous copper sulfate, stopper and shake, at intervals, for several hours and allow to stand over night. Distill through a fractionating column fitted with a dry, water-cooled condenser, into a closed receiver. Protect the distillate receiver with a calcium chloride drying tube. Discard the first 25 ml. of the distillate, collect about 1500 ml. of the distillate in a brown glass bottle, fitted with a carefully ground glass stopper, and store in a desiccator. (A desiccator of suitable size, provided with a low form, glass, bell jar, as a cover, is satisfactory.)

4.4.1.3 *Sodium iodide.*—Place 150 ml. of distilled water in a porcelain evaporating dish approximately 305 mm. in diameter. Add 450 gm. of sodium iodide slowly to the water in the dish while heating with constant stirring, and heat and stir until the sodium iodide is in solution. When dissolved, add dilute hydrochloric acid until faintly acid to litmus. Transfer the solution to a beaker and cool to 0° to 10° C. Filter off the crystals on a Buchner funnel. Apply suction to remove water. Transfer the crystals to an evaporating dish and heat on a hot plate, with stirring, until the crystals are white and dry. Place in a bottle and store in an oven at 100° to 110° C.

4.4.2 *Phosgene determination.* — Dry the 50-ml. bottle, shown on figure 1, in an oven at 105° ± 5° C. and cool in a desiccator. Place

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the bottle in the metal carrier and suspend the assembly in a Dewar cylinder previously charged with a mixture of carbon dioxide snow and acetone. Discharge about 10 ml. of phosgene from the steel sampling cylinder into the bottle. The sampling cylinder, containing the phosgene, should be previously cooled for at least 2 hours in a bath of ice and water, then removed and shaken. Place the cylinder, on a support, in an inclined position so that the liquid phase occupies the lower end of the cylinder. The vapor in the cylinder should not be vented before the withdrawal of the sample. Swirl the phosgene to wash the inside of the 50-ml. bottle and discard. Immediately replace the bottle in the metal carrier suspended in the Dewar cylinder and fill with phosgene from the steel cylinder to near the shoulder. Close with the capsule holding a tared glass bulb (dried in an oven at 100° to 110° C. and cooled in a desiccator before taring) in the position shown in the figure. Place an amount of carbon dioxide snow in the capsule which will cover most of the bulb, leaving sufficient surface uncovered to allow observation of the contents of the bulb. When about 0.5 gm. of sample has been drawn into the bulb, cover the capsule with an insulating pad of absorbent cotton or cloth to protect the hand, remove and hold at about a 45 degree angle with the stem pointing downward. Bring the stem near a gas flame to warm it slightly. When the liquid leaves the capillary, seal quickly in the flame, with the bulb still in the capsule of carbon dioxide snow. Allow the bulb to come to room temperature in a desiccator and weigh to the nearest 0.1 mg. Transfer approximately 5 gm. of dried sodium iodide (see 4.4.1.3) while still hot into a 300- or 500-ml. flask provided with a glass stopper that does not leak. (The tightness of this closure should be determined before the flask is used.) The flask should be dried at 100° to 110° C. and still be warm after removal from the oven. Grease the stopper with pure petroleum jelly. Add approximately 30 ml. of dry acetone (see 4.4.1.2). Place the bulb containing the phosgene in the flask, stopper tightly, cool in an ice bath to well below room temperature, and shake to break the bulb. Continue to shake for 3 minutes to absorb the phosgene, cool in the ice bath, remove the stopper, and wash down the stopper and neck of the flask with a small amount of distilled water. Titrate with standard 0.1*N* sodium thiosulfate solution, which has been made neutral to phenolphthalein, to the disappearance of the iodine color. When near the end point crush the fragments of the glass capillary with a glass rod and rinse the rod with distilled water. Similarly titrate a prepared blank, containing a quantity of acetone and sodium iodide solution equal to that used in the determination, with sodium thiosulfate solution. Reserve the solutions from the determination and blank for the acidity test described in 4.4.3.

$$\text{Percent phosgene} = \frac{(A - B) \times N \times 4.946}{\text{weight of sample}} - (C \times 1.396)$$

where:

A = ml. $Na_2S_2O_3$ used in the determination.
B = ml. $Na_2S_2O_3$ used in blank.
C = percent free chlorine.
N = normality of $Na_2S_2O_3$ solution.

4.4.3 *Acidity.*—Add 0.5 ml. of 0.2-percent phenolphthalein indicator to the flasks, containing the sample and the blank respectively, from the phosgene determination described in 4.4.2. Titrate the contents of both flasks with 0.1*N* sodium hydroxide to the same faint pink color.

$$\text{Percent acidity as HCl} = \frac{(\text{ml. NaOH} - \text{ml. blank}) \times N \times 3.65}{\text{weight of sample}}$$

where:

N = normality of NaOH solution.

4.4.4 *Residue on evaporation.* — Introduce into a tared Erlenmeyer flask of 50-ml. capacity, with a calibration mark at 35 ml., sufficient phosgene to fill the flask to the 35-ml. mark (50 gm.) Allow the phosgene to evaporate in a vacuum desiccator at atmos-

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pheric pressure. When the liquid has evaporated, apply a vacuum of 200 mm., or less for one-half hour, then admit dried air to the desiccator. Repeat until the odor of phosgene cannot be detected in the flask, then remove the flask and weigh.

$$\text{Percent residue} = \frac{\text{weight of residue} \times 100}{\text{weight of sample (50 gm.)}}$$

4.4.5 *Free chlorine.*

4.4.5.1 *Free chlorine in grade A phosgene.*—Place 1.0 to 1.5 gm. of sample in a thin glass bulb as described in 4.4.2. Place the bulb in a 750- or 1,000-ml., heavy-walled, glass-stoppered flask, containing 100 ml. of cold 3-percent aqueous potassium iodide solution. Stopper the flask and break the bulb by shaking the flask. Continue to shake the flask for at least 3 minutes. Cool the flask in an ice bath before removing the stopper, and titrate immediately with 0.05*N* sodium thiosulfate, using starch as an indicator.

$$\text{Percent free chlorine} = \frac{\text{ml. } Na_2S_2O_3 \times N \times 3.55}{\text{weight of sample}}$$

where:

N = normality of $Na_2S_2O_3$ solution.

4.4.5.2 *Free chlorine in grade B phosgene.*—Place 1.0 to 1.5 gm. of sample in a thin glass bulb as described in 4.4.2. Place the bulb in a 750- or 1,000-ml., heavy-walled, glass-stoppered flask containing 200 ml. of distilled water. Stopper the flask and break the bulb by shaking the flask. Continue to shake the flask for at least 3 minutes. Cool the flask in an ice bath before removing the stopper, add 5 gm. of solid potassium iodide, swirl the flask to mix the contents, and titrate immediately with 0.05*N* sodium thiosulfate, using starch as an indicator.

$$\text{Percent free chlorine} = \frac{\text{ml. } Na_2S_2O_3 \times N \times 3.55}{\text{weight of sample}}$$

where:

N = normality of $Na_2S_2O_3$ solution.

5. PREPARATION FOR DELIVERY

5.1 Packing.—The phosgene shall be packed in a 1-ton steel container, conforming to Specification 96-31-3 or MIL-C-3250. Each 1-ton container shall contain 1,900 ± 10 pounds (net weight) of phosgene. Before filling, the inside of the container shall be examined for the presence of foreign material and cleaned and dried if necessary.

5.2 Marking.—In addition to any special marking required by the contract or order and by Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles, etc., all shipping containers for the Army shall be marked in accordance with Specification 100–2, except that the standard nomenclature shall be in a code to be furnished by the Chief Chemical Officer; for the Air Force, in accordance with Specification 94–40645; for the Navy, in accordance with the Navy Shipment Marking Handbook.

6. NOTES

6.1 Ordering data.—Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Grade of phosgene required (see 1.2).

c. Type of packing required (see 5.1).

6.2 Alternate sampling method.—If desired a continuous sampling method may be used. The following method is suggested: Divert a small portion of the condensed phosgene into a suitable receiver. Agitate the diverted phosgene and collect the composite sample, for any desired amount of production, in a steel sampling cylinder. Label the cylinder to identify the contents

6.3 Inspection.—Inspection will not necessarily include all specification requirements; however, the omission of inspection does not constitute a waiver of these requirements.

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6.4 **Sampling.**—If the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.5 Copies of A.C.S. specifications may be obtained from the American Chemical Society, Washington 6, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

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Other interest:

Army—T

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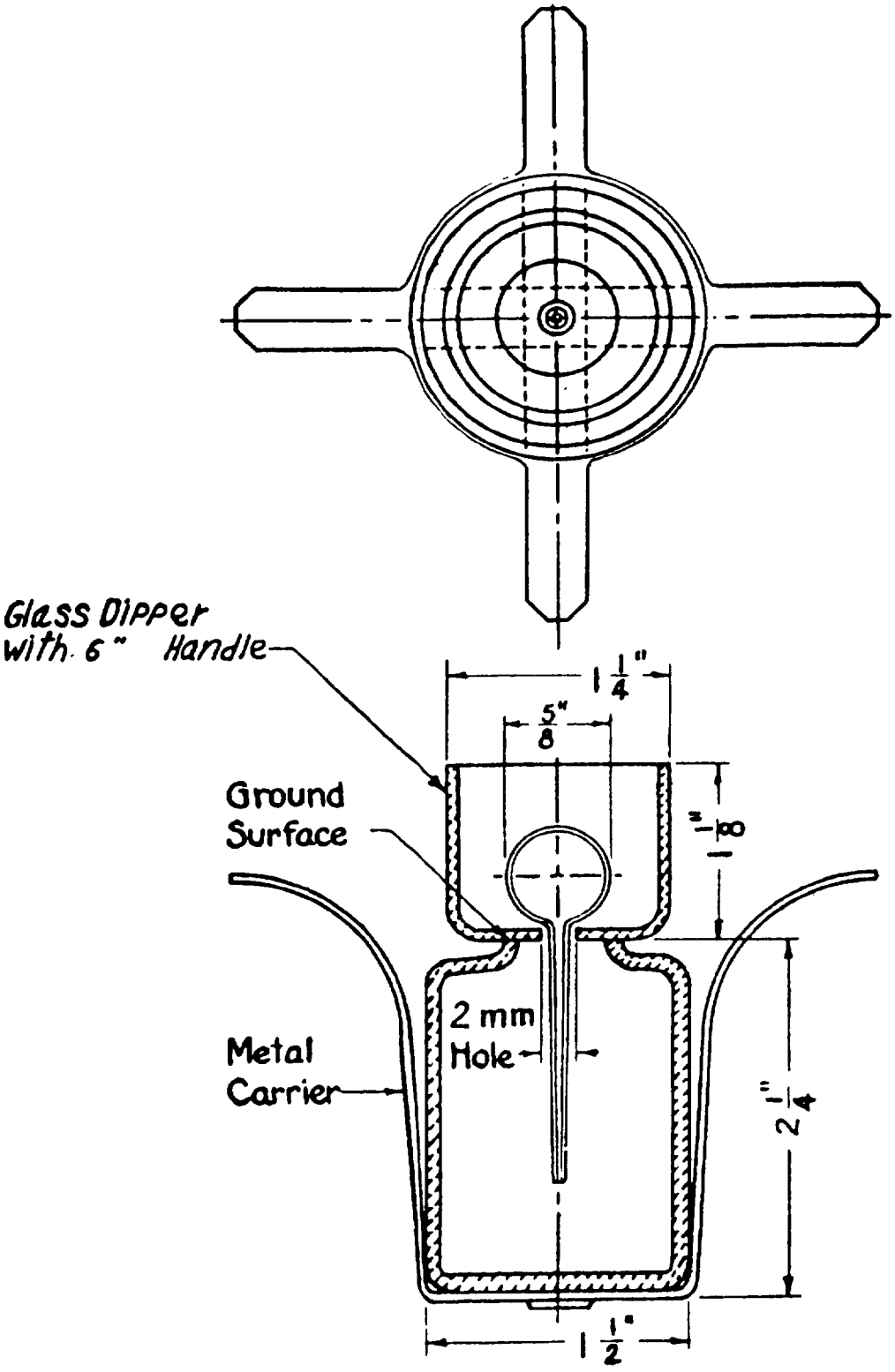


FIGURE 1.—*Sampling apparatus for liquefied gases.*

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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